Tianlong Wang

PhD candidate | Zhejiang University | Geohazards, hydrology and artificial intelligence

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Academic homepage | Google Scholar | ORCID | ResearchGate | LinkedIn | GitHub

Research profile

I develop physics-informed and explainable machine-learning methods for debris-flow, landslide and hydrological prediction. My research combines hydrological reasoning, residual learning, ensemble optimization and uncertainty analysis, with an emphasis on small catchments and sparse observations. I expect to complete my PhD in June 2027 and seek international postdoctoral opportunities from July 2027 in hydrology, geohazards, geotechnics and related AI applications.

Research outputs: 14 published or accepted first- or corresponding-author publications (13 journal articles and 1 conference paper); 6 manuscripts under review; 8 invention-patent records (3 grant publications and 5 application publications); 6 open-source research software projects.

Lead investigator on 6 research and innovation projects, including 4 funded awards with recorded support totalling RMB 136,000; contributor to 7 collaborative projects; separate CSC joint-training support of RMB 300,000. The record includes 10 honors and scholarship entries and 12 competition-award entries; reviewer for 16 journals and member of 3 professional societies.

Education and joint doctoral training

Zhejiang University | PhD candidate | Sep 2022 - Jun 2027 (expected)

Ocean College; Ocean Technology and Engineering; Supervisor: Hongyue Sun

Nanyang Technological University | Joint PhD Student | Sep 2024 - Sep 2026

School of Civil and Environmental Engineering; China Scholarship Council sponsorship; Supervisor: Chu Jian

Research projects and funding

Investigator-led research and innovation projects 6

1. **Spatiotemporal Dynamic Forecasting and Interpretive Analysis System for Debris Flows in Small Watersheds**. Young Elite Scientists Sponsorship Program Doctoral Student Special Plan, China Association for Science and Technology; PI; January 2025 — December 2026. Project ID: 156-O-170-0000603-2 Funding: RMB 40,000.
2. **Research on Intelligent Debris Flow Forecasting Methods for Small Watersheds Based on Multi-scale Hydrodynamic Coupling Mechanisms**. Seed Fund Cultivation Project of Ocean College, Zhejiang University; PI; June 2025 — June 2027. Grant No. 2025BS004 Funding: RMB 50,000.
3. **Spatiotemporal Dynamic Forecasting and Interpretive Analysis System for Debris Flows in Small Watersheds**. Outstanding Doctoral Dissertation Award Cultivation Grant, Zhejiang University; PI; 2025–2026. Grant No. 825011A · RMB 36,000.
4. SLOPE GUARD: Integrated Landslide Monitoring and Drainage Solution Provider; China International College Students' Innovation Competition, Singapore Regional project; RMB 10,000; project 00006; completed with distinction; 2025.06–2025.11
5. Application of magnetic fluids to visualized fracture seepage and grouting-based seepage control; university research project; completed with distinction
6. Energy dissipation, load reduction and reliability of flexible cushion layers for gravity retaining walls; university research project; completed with distinction

Collaborative research projects 7

1. **Early Forecasting and Disaster Control Methods for Debris Flows in Small Watersheds Based on Hydrodynamic Analysis**. National Natural Science Foundation of China, Key Program; Core team member; January 2023 — December 2027. Grant No. 42230702.

2. **Research on Deformation Prediction Models and Instability Criterion Methods for Reservoir Bank Slopes Based on Copula Theory**. National Natural Science Foundation of China, Young Scientists Fund; Core team member; January 2019 — December 2021. Grant No. 51809151. Completed.
3. Geomechanical mechanisms of collapse and landslide hazards on steep slopes under climate change; CAST Young Elite Scientists Sponsorship Program; Core team member.
4. Novel drainage treatment technologies for soft soil; major provincial commissioned research project; completed with distinction; Core team member.
5. Smart monitoring system for digitally instrumented towers; provincial student innovation project; completed with distinction; Core team member.
6. Innovative methods and technologies for slope protection and treatment; provincial student innovation project; completed with distinction; Core team member.
7. Integrated space-air-ground landslide monitoring and siphon-drainage mitigation system; provincial student innovation project; completed with distinction; Core team member.

Joint doctoral training scholarship 1

China Scholarship Council, National Construction of High-Level Universities Program; Sep 2024 - Sep 2026; RMB 300,000; grant 202406320358; scholarship recipient.

Manuscripts under review 6

1. **Tianlong Wang**, Jian Chu, Hao Yang, Hongyue Sun*. Physics-guided residual stacking for short-lead runoff and stage forecasting in the Fengshuwan catchment, China. *Journal of Hydrology: Regional Studies*. Under review following revision.
2. **Tianlong Wang**, Jian Chu, Jingkun Su, Runting Fan, Hao Yang, Hongyue Sun*. Hydrological readiness forecasting for debris flow early warning in a mountain catchment with sparse event records. *Bulletin of Engineering Geology and the Environment*. Under review.
3. **Tianlong Wang**, Jian Chu, Hao Yang, Hongyue Sun*. Robust state estimation and hydrology-informed residual learning for landslide velocity forecasting with prequential prediction intervals. *Georisk: Assessment and Management of Risk for Engineered Systems and Geohazards*. Under review.
4. **Tianlong Wang**, Jian Chu, Yanhui Li, Hao Yang, Hongyue Sun*. Sediment-source information improves small-sample prediction of maximum recorded debris-flow volume with a pretrained tabular model. *Stochastic Environmental Research and Risk Assessment*. Under review following revision.
5. Hao Yang, **Tianlong Wang***, Haijun Qiu, Nikita Igorevich Fomin, Shuoting Xiao, Pengyue Zhao. An explainable and metaheuristic-optimized machine learning framework for spatial debris flow risk prediction: Application to the northern China-Pakistan Economic Corridor. *Stochastic Environmental Research and Risk Assessment*. Under review following second revision.
6. Yanhui Li, **Tianlong Wang***, Xiaorui Tao. DynaStack: An explainable and robust ensemble framework for trend-based stock price prediction and strategy optimization. *Journal of Forecasting*. Under review following second revision.

Published and accepted papers (first or corresponding author) 14

1. **Tianlong Wang**, Jian Chu, Hao Yang, Hongyue Sun*. Basin water storage degree for risk-informed debris-flow warning support: A forward-looking proof of concept in a small mountain catchment. *International Journal of Disaster Risk Reduction*, 146, 106496 (2026). Article DOI | Data | Code
2. **Tianlong Wang***, H. Yang. DOME: A Dynamic Optimization Meta-Ensemble Framework for Debris-Flow Susceptibility Assessment with SHAP-Based Interpretation. *Natural Hazards, 122, Article 545*, 2026. https://doi.org/10.1007/s11069-026-08307-2
3. **Tianlong Wang**, H. Yang, H. Sun*. Intelligent Prediction of Post-Seismic Debris Flow Source Volumes: A Physics-Informed Generative Deep Learning Framework. *Georisk: Assessment and Management of Risk for Engineered Systems and Geohazards, 1–29*, 2026. https://doi.org/10.1080/17499518.2026.2651228

4. H. Yang, **Tianlong Wang***, N. I. Fomin, S. Xiao, L. Liu. An integrated IKOA-CNN-BiGRU-Attention framework with SHAP explainability for high-precision debris flow hazard prediction in the Nujiang river basin, China. *PLOS ONE, 20(6), e0326587*, 2025. https://doi.org/10.1371/journal.pone.0326587
5. **Tianlong Wang**, Q. Ge, T. Ma, et al.. A novel method for predicting debris flow hazard: a multi-strategy fusion approach based on the light gradient boosting machine framework. *Stochastic Environmental Research and Risk Assessment, 39, 4867–4890*, 2025. https://doi.org/10.1007/s00477-025-02955-9
6. **Tianlong Wang**, K. Zhang, Z. Liu, T. Ma, R. Luo, H. Chen, X. Wang, W. Ge, H. Sun*. Prediction and explanation of debris flow velocity based on multi-strategy fusion Stacking ensemble learning model. *Journal of Hydrology, 638, 131347*, 2024. https://doi.org/10.1016/j.jhydrol.2024.131347
7. **Tianlong Wang**, R. Luo, T. Ma, et al.. Study and verification on an improved comprehensive prediction model of landslide displacement. *Bulletin of Engineering Geology and the Environment, 83, Article 90*, 2024. https://doi.org/10.1007/s10064-024-03581-5
8. Z. Chu, S. Yu, X. Li, X. Lu, X. Wang, R. Luo, **Tianlong Wang***. Indoor Model Test of Negative Pressure Drainage Method for Treatment of Engineering Mud. *Journal of Changjiang River Scientific Research Institute, 41(9), 114–122, 129. (In Chinese)*, 2024. https://doi.org/10.11988/ckyyb.20230665
9. **Tianlong Wang**, D. Peng, X. Wang, B. Wu, R. Luo, Z. Chu, H. Sun*. Study on wavelet multi-scale analysis and prediction of landslide groundwater. *Journal of Hydroinformatics, 26(1), 237–254*, 2024. https://doi.org/10.2166/hydro.2023.299
10. **Tianlong Wang**, Xiaorui Tao, Dong-kun Wu, Haotian Feng. Determining the location of River Dam Group based on set cover Model: A case study of Zambezi River Basin. *Scientific Research and Reviews, 14, 122*, 2021. https://doi.org/10.28933/srr-2021-03-1005
11. **Tianlong Wang**, J. Wang, X. Shao. Carbon emission forecast of China Transport Industry Based on Grey - Markov. *International Conference on Computing and Big Data (ICCBD)*, 2021. https://doi.org/10.1109/ICCBD53833.2021.00192
12. **Tianlong Wang**, H. Zhang, X. Tao, W. Chu. Research on Prediction and Optimization of Slope Deformation in Mining Area. *Environment, Resource and Ecology Journal, 4, 66–69*, 2020. https://doi.org/10.23977/erej.2020.040110
13. **Tianlong Wang**, Hongyan Bao, Hairui Zhang. Optimization of Task Pricing Based on Multiple Regression Analysis and Game Theory. *Journal of eSciences, 3, 9*, 2020. https://doi.org/10.28933/esciences-2020-01-1805
14. **Tianlong Wang**, Xiaorui Tao, Jiamei Zhang, Yulei Li. Predicting Traffic Congestion Time Based on Kalman Filter Algorithm. *Advances in Research and Reviews, 1, 7*, 2020. https://doi.org/10.28933/arr-2020-06-2205

Invention patents and applications 8

Grant publications 3

1. J. Liu, Z. Li, **Tianlong Wang**, et al.. Eccentric Hook-shaped Expansible Pre-stressed Anchoring Structure and Its Anchoring Method. China; CN113718767B; 2023. Patent record
2. J. Liu, Q. Shi, Z. Wei, S. Li, L. Tang, **Tianlong Wang**, et al.. Device and Method for Obtaining the Position of Liquid Displacement Surface in Rock Cracks. China; CN111595550B; 2022. Patent record
3. J. Liu, Q. Shi, Z. Wei, S. Li, L. Tang, **Tianlong Wang**, et al.. Pile Structure and Construction Method in Coral Sand Environment. China; CN110886291B; 2021. Patent record

Published applications 5

1. H. Sun, X. Wang, **Tianlong Wang**, R. Tang, R. Luo. Monitoring and Early Warning Method and System for Rainfall-induced Landslides. Japan; application 2024-017304; JP2025113090A; 2025. Patent record
2. X. Wang, H. Sun, S. Shao, R. Luo, **Tianlong Wang**. Early Warning and Mitigation Method for Debris Flow Disasters in Small Watersheds. China; CN118262481A; 2024. Patent record
3. X. Wang, H. Sun, **Tianlong Wang**, R. Tang, R. Luo. Monitoring and Early Warning Method and System for Rainfall-induced Landslides. China; CN117854239A; 2024. Patent record
4. **Tianlong Wang**, H. Sun, Z. Chu, R. Luo, X. Wang. Landslide Displacement Prediction Method Based on IVMD-IAO-BiLSTM. China; CN116757323A; 2023. Patent record

5. J. Liu, Z. Li, **Tianlong Wang**, et al.. Method for Repairing Dam Crack Defects Using Magnetic Mortar. China; CN114396022A; 2022. Patent record

Open-source research software

1. **DOME**. Dynamic ensemble selection and optimization for regional debris-flow susceptibility, with SHAP interpretation. Repository
2. **TabPFNv2.5-DebrisFlow**. Small-sample debris-flow volume prediction using a pretrained tabular model and sediment-source information. Repository
3. **RSRL-HU**. One-day-ahead landslide velocity forecasts with hydrology-informed residual learning and sequential intervals. Repository
4. **Hydrological Readiness Forecasting**. Three-day probabilistic forecasts of warning-relevant hydrological states, evaluated chronologically. Repository
5. **BWSD**. Reproducible apparent-storage assessment with sequential reference updating, sensitivity analysis and uncertainty propagation. Repository
6. **HydroPIML-RS**. Physics-guided residual stacking for short-lead runoff and water-stage prediction. Repository

Academic service

Journal reviewer: Journal of Hydrology; Acta Geotechnica; Bulletin of Engineering Geology and the Environment; Stochastic Environmental Research and Risk Assessment; Powder Technology; PLOS ONE; Applied Intelligence; Earth Science Informatics; Groundwater for Sustainable Development; International Journal of Computational Intelligence Systems; Journal of Rock Mechanics and Geotechnical Engineering; Air, Soil and Water Research; Discover Geoscience; Earth Sciences; Array; International Journal of Interactive Multimedia and Artificial Intelligence.

Professional memberships (3): Chinese Society for Rock Mechanics and Engineering; Geological Society of China; China Civil Engineering Society.

Honors scholarships and competition awards 22 entries

Honors and scholarships

1. National Scholarship; 2025.
2. Chinese Government Scholarship, China Scholarship Council; 2024.
3. China Harbour Engineering Company Scholarship; 2023.
4. Outstanding Graduate, Chinese Society for Rock Mechanics and Engineering; one of 100 recipients nationwide; 2022.
5. Outstanding Graduate, university level; 2022.
6. Yunneng Electric Power Innovation Scholarship; 2022.
7. National Scholarship; 2021.
8. Wang Shuopeng Scholarship; 2020.
9. First-Class Academic Scholarship, four consecutive years; 2019–2022.
10. National Encouragement Scholarship, three consecutive years; 2019–2021.

Competition awards

1. Silver Award, China International College Students' Innovation Competition, Singapore Regional; 2025.
2. Silver Award, Zhejiang International College Students' Innovation Competition (Internet+); 2024.
3. Bronze Award, 14th Challenge Cup Zhejiang Undergraduate Entrepreneurship Plan Competition; 2024.
4. Silver Award, 15th Dandelion University Student Entrepreneurship Competition, Zhejiang University; 2023.
5. Honorable Mention, Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling (MCM/ICM); 2022.
6. Second Prize, national final, 14th National University Student Social Practice and Science Contest on Energy Saving and Emission Reduction; 2021.
7. First Prize, Southern Region; Third Prize, National, 8th National Financial and Securities Investment Simulation Training Contest; 2021.
8. Grand Prize, 3rd National College Student Structural Design Information Technology Competition; 2021.
9. Third Prize, Hubei, 13th Challenge Cup National Undergraduate Curricular Academic Science and Technology Works Competition; 2021.
10. Honorable Mention, Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling (MCM/ICM); 2021.
11. Second Prize, Hubei, Contemporary Undergraduate Mathematical Contest in Modeling; 2020.
12. Third Prize, Hubei, 5th National Construction College Construction Technology Application Skills Competition; 2020.

Additional experience

1. Government affairs intern, Science and Technology Exchange Division, China Centre for International Science and Technology Exchange, CAST. 2023.07–2023.09
2. Student fitness club coordinator, Ocean College, Zhejiang University; team champion in the university fitness and bodybuilding competition, 2023. 2022.06–2024.06
3. Author of the published poetry collection Suganniran. 2025
4. Volunteer teacher, summer education programme, Songbai, Shennongjia, Hubei. 2019.07–2019.08
5. Construction site intern, Yujing Tiandi project, Kunfa Construction Company. 2019.06–2019.07